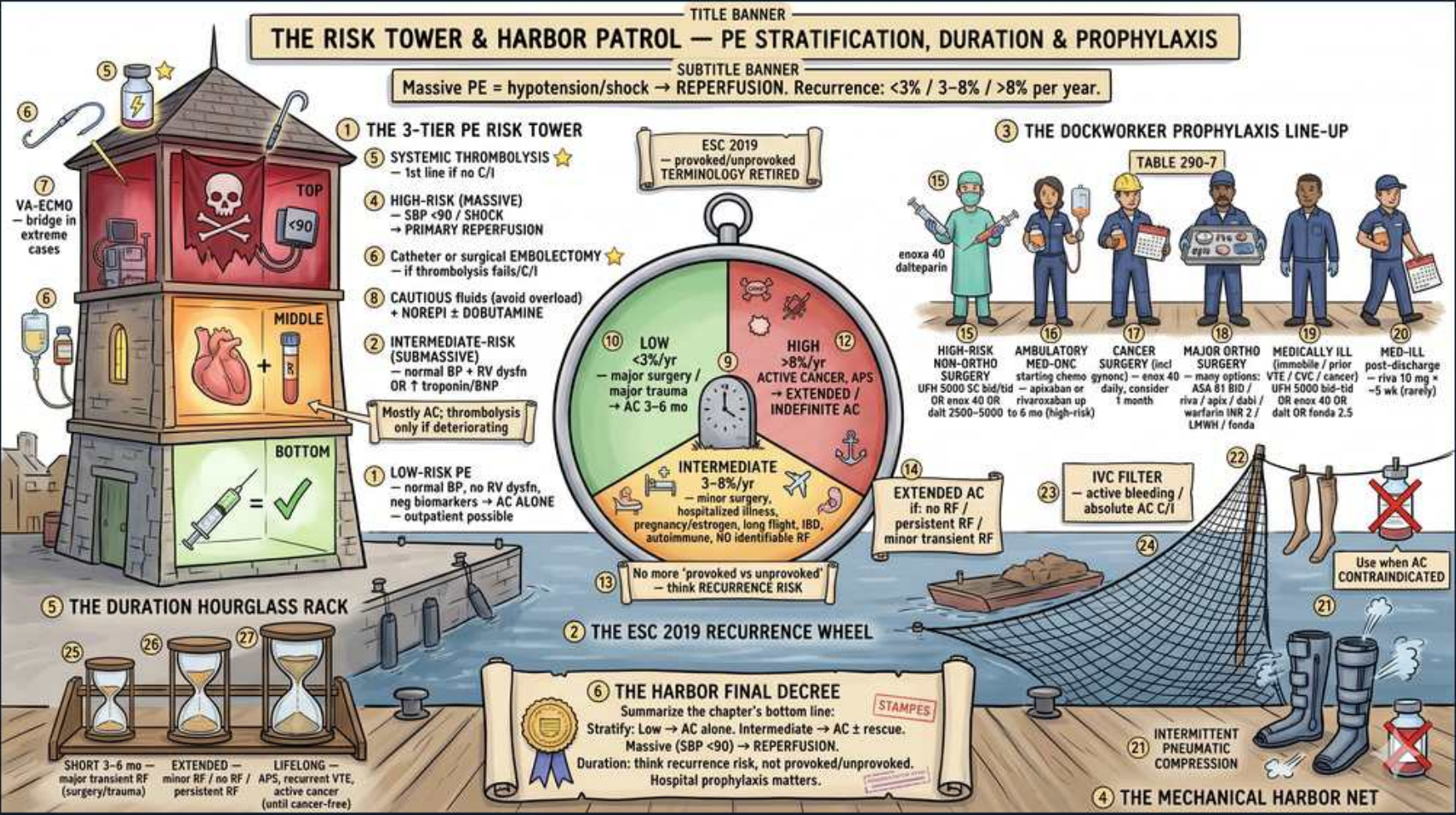


VTE — Risk Stratification, Treatment Duration & Prophylaxis



1. High-risk (massive) PE

WHAT YOU SEE

Top of risk tower with skull, labeled high-risk/massive

WHAT IT ENCODES

High-risk (massive) PE is defined hemodynamically by SBP <90 mmHg for ≥ 15 min, a drop ≥ 40 mmHg, or need for pressors — i.e. obstructive shock, NOT by clot size. Treatment is immediate systemic thrombolysis (alteplase/tPA 100 mg IV over 2 h) plus heparin if there is no contraindication; mortality runs >25-50%. Think 'reperfusion = REPERFUSION' for the unstable patient.

BOARD TRAP

WRONG: 'order a CTPA first to confirm the clot before treating.' In a crashing hypotensive patient you do not delay for confirmatory imaging — give empiric anticoagulation $\pm$ thrombolysis (or bedside echo showing RV strain) because the patient can arrest in the scanner. CTPA is for stable patients.

2. Intermediate-risk (submassive) PE

WHAT YOU SEE

Middle tier: normal BP but RV strain, positive troponin/BNP

WHAT IT ENCODES

Intermediate (submassive) PE = NORMOTENSIVE patient WITH evidence of RV dysfunction (echo/CT RV dilation, RV/LV ratio >0.9) and/or positive cardiac biomarkers (troponin, BNP). Subdivided into intermediate-high (both RV strain AND positive troponin) vs intermediate-low. Management is anticoagulation with close monitoring; consider rescue thrombolysis only if they deteriorate to hemodynamic instability.

BOARD TRAP

WRONG: 'give full-dose systemic thrombolysis to every submassive PE.' The PEITHO trial showed routine thrombolysis in stable intermediate-risk PE reduced decompensation but caused significantly more major bleeding and intracranial hemorrhage with no mortality benefit. Anticoagulate and watch; reserve lysis for those who decompensate.

3. Low-risk PE

WHAT YOU SEE

Bottom tier: normal BP, no RV strain, negative biomarkers

WHAT IT ENCODES

Low-risk PE = normotensive, NO RV strain on imaging, and negative biomarkers. A low PESI/sPESI score (sPESI = 0) identifies patients with 30-day mortality ~1% who may be managed as outpatients on a DOAC. Begin anticoagulation immediately; admission is not mandatory if home circumstances are adequate.

BOARD TRAP

WRONG: 'admit every PE patient for IV heparin.' Validated low-risk patients (sPESI 0, no RV dysfunction, good social support, no renal/bleeding contraindication) can be discharged on a DOAC such as rivaroxaban or apixaban — over-admitting low-risk PE is not the boards answer.

4. Catheter / surgical embolectomy

WHAT YOU SEE

Note: catheter-directed or surgical embolectomy if thrombolysis fails/contraindicated

WHAT IT ENCODES

When high-risk PE has FAILED systemic thrombolysis or thrombolysis is CONTRAINDICATED (recent surgery, active bleeding, hemorrhagic stroke, intracranial neoplasm), the next step is mechanical clot removal: catheter-directed thrombolysis/thrombectomy or open surgical embolectomy. Catheter-directed therapy uses much lower local lytic doses to cut bleeding risk.

BOARD TRAP

WRONG: 'give a second full systemic lytic dose if the first fails.' Repeated systemic thrombolysis dramatically escalates intracranial hemorrhage risk; the correct escalation is catheter-directed or surgical embolectomy (or ECMO bridge).

5. VA-ECMO bridge

WHAT YOU SEE

Top icon VA-ECMO bridge cases

WHAT IT ENCODES

Veno-arterial ECMO provides temporary cardiopulmonary support to bridge refractory high-risk PE or PE-associated cardiac arrest to definitive therapy (embolectomy/clot resolution). It oxygenates and offloads the failing right ventricle while the obstruction is addressed. Reserved for the truly crashing patient at experienced centers.

BOARD TRAP

WRONG: 'ECMO is a definitive treatment for PE.' ECMO is only a BRIDGE — it buys time for thrombolysis, embolectomy, or autolysis; the clot still must be removed or dissolved.

6. Recurrence wheel: provoked low risk

WHAT YOU SEE

Wheel green slice: provoked by major transient risk <3% recurrence

WHAT IT ENCODES

VTE provoked by a MAJOR TRANSIENT risk factor (surgery within 3 months under general anesthesia, major trauma, hospitalization with immobility, cesarean) carries low annual recurrence (~3%) once the trigger resolves. Standard duration is exactly 3 months of anticoagulation, then stop. The dominant transient driver is what makes it 'low recurrence.'

BOARD TRAP

WRONG: 'extend a surgery-provoked DVT indefinitely to be safe.' Indefinite anticoagulation in a clearly provoked-by-transient-factor VTE exposes the patient to bleeding with no net benefit — 3 months is the answer when the provoking factor is gone.

7. Recurrence wheel: unprovoked / high

WHAT YOU SEE

Wheel red slice: unprovoked / active cancer / APS = high recurrence

WHAT IT ENCODES

Unprovoked VTE, active cancer, or antiphospholipid syndrome (APS) confer HIGH recurrence (>8%/yr off treatment) → extended/indefinite anticoagulation if bleeding risk is acceptable. Cancer-associated VTE is treated with a DOAC (apixaban/rivaroxaban/edoxaban) or LMWH continued while cancer is active; APS is treated with warfarin (DOACs are inferior, especially triple-positive APS).

BOARD TRAP

WRONG: 'treat APS-associated VTE with a DOAC for convenience.' The TRAPS trial showed rivaroxaban increased arterial thrombotic events in triple-positive APS — warfarin (INR 2-3) is the correct agent for APS, not a DOAC.

8. Short duration 3 months

WHAT YOU SEE

Hourglass: short 3 months for provoked (surgery/trauma)

WHAT IT ENCODES

SHORT course = 3 months total, for VTE provoked by a major transient factor (surgery, trauma, immobilization). Three months is the minimum that fully treats the index clot; stopping earlier raises early recurrence. After completion, no routine thrombophilia testing is needed for a clearly provoked event.

BOARD TRAP

WRONG: 'stop anticoagulation at 6 weeks because the leg looks better.' Symptomatic improvement does not equal clot resolution/endothelial healing — the minimum is 3 months even for provoked VTE.

9. Extended duration

WHAT YOU SEE

Hourglass: extended >3 months, minor RF or first unprovoked

WHAT IT ENCODES

EXTENDED (>3 months, reassessed periodically) is for a first UNPROVOKED VTE or VTE with a persistent/minor risk factor, when bleeding risk is low-moderate. Low-dose DOAC (apixaban 2.5 mg BID or rivaroxaban 10 mg daily) after the initial treatment phase reduces recurrence with little excess bleeding. Use a bleeding tool (e.g. VTE-BLEED/HAS-BLED context) plus patient preference to decide.

BOARD TRAP

WRONG: 'a first unprovoked VTE always gets exactly 3 months then stop.' Unprovoked VTE has high recurrence; guidelines favor EXTENDED anticoagulation unless bleeding risk is high — the reflex 3-month stop is the trap.

10. Lifelong duration

WHAT YOU SEE

Hourglass: lifelong for recurrent VTE / active cancer

WHAT IT ENCODES

LIFELONG/indefinite anticoagulation is for RECURRENT unprovoked VTE, active cancer (while cancer persists), or APS — the highest-recurrence groups. The decision is revisited at intervals balancing thrombotic vs bleeding risk. 'Indefinite' does not mean 'never reassess.'

BOARD TRAP

WRONG: 'add an IVC filter to lifelong anticoagulation for extra protection in recurrent VTE.' Adding a filter on top of effective anticoagulation provides no benefit and adds filter complications (thrombosis, migration) — filters are only for when anticoagulation is contraindicated.

11. Prophylaxis line-up

WHAT YOU SEE

Dockworkers in line labeled prophylaxis (Padua/Caprini)

WHAT IT ENCODES

Every hospitalized adult should be VTE-risk-stratified at admission. Use the PADUA score for medical inpatients (≥4 = high risk → prophylaxis) and the CAPRINI score for surgical patients (escalating prophylaxis by tier). Risk assessment, not blanket dosing, drives whether pharmacologic prophylaxis is started.

BOARD TRAP

WRONG: 'give pharmacologic prophylaxis to every admitted patient.' Low-risk medical inpatients (Padua <4) and low-risk surgical patients do NOT need pharmacologic prophylaxis — universal heparin causes unnecessary bleeding. Stratify first.

12. Pharmacologic prophylaxis (LMWH)

WHAT YOU SEE

High-risk worker labeled LMWH/UFH/DOAC prophylaxis

WHAT IT ENCODES

Pharmacologic prophylaxis for high-risk inpatients: LMWH (enoxaparin 40 mg SC daily, or 30 mg BID in trauma/ortho), low-dose unfractionated heparin (5000 units SC q8-12h, preferred in severe renal failure CrCl <30), or fondaparinux (HIT history). Extended post-discharge prophylaxis is standard after total hip/knee arthroplasty (LMWH or a DOAC) and high-risk cancer surgery.

BOARD TRAP

WRONG: 'use LMWH for prophylaxis in a patient with CrCl 20.' LMWH accumulates in severe renal impairment (CrCl <30) → bleeding; switch to unfractionated heparin (or dose-reduce/monitor anti-Xa). UFH is renally safe.

13. IVC filter when AC contraindicated

WHAT YOU SEE

Net/filter labeled use when anticoagulation contraindicated

WHAT IT ENCODES

An IVC filter is indicated ONLY when anticoagulation is absolutely contraindicated in a patient with acute proximal DVT/PE (e.g. active major bleeding, recent intracranial hemorrhage), or for recurrent PE despite therapeutic anticoagulation. Use a RETRIEVABLE filter and remove it once anticoagulation can be resumed; it prevents PE but does not treat the clot or reduce DVT.

BOARD TRAP

WRONG: 'place a permanent IVC filter instead of anticoagulation in a stable PE patient with no bleeding.' Filters are not a substitute for anticoagulation and actually increase the risk of recurrent DVT long-term — they are a niche bailout for AC contraindication, not first-line therapy or prophylaxis.

14. Mechanical prophylaxis (IPC)

WHAT YOU SEE

Intermittent pneumatic compression sleeves / stockings

WHAT IT ENCODES

Mechanical prophylaxis — intermittent pneumatic compression (IPC) and graduated compression stockings — is used when pharmacologic prophylaxis is CONTRAINDICATED (active bleeding, severe thrombocytopenia, high bleeding risk) or as an adjunct in very-high-risk patients. IPC is preferred over stockings (CLOTS trials showed stockings alone are weak/ineffective and harmful in stroke).

BOARD TRAP

WRONG: 'graduated compression stockings are good VTE prophylaxis after stroke.' The CLOTS-1 trial found thigh-length stockings did not reduce DVT after stroke and increased skin breakdown — IPC is the mechanical method of choice when drugs are contraindicated.
